

१. Write short note on the following:

a. Status of irrigation development in Nepal

b. Crop water requirements calculation by penman methods

c. Types of irrigation

Answer:

a. Status of irrigation development in Nepal:

- The history of irrigation in Nepal before १९२२ indicate that irrigation was developed, operated and maintained by farmers known as Farmers Managed Irrigation System (FMIS).
- Out of total potential irrigable area, ७६% lies in Terai region of Nepal.
- Based on the **Irrigation Master Plan २०१९ (Updated २०२४)** published by the **Department of Water Resources and Irrigation (DWRI)**, Out of total geographical area of Nepal i.e., १४.७ million ha, २.९७ million ha is the agricultural land. Master Plan identifies **about २.५ million hectares** as potentially irrigable. By २०२४, irrigation infrastructure had reached approximately **१.५ million hectares**, but only about **one-third** of this area receives reliable **year-round irrigation**.
- From १९२२ to १९५७, Government made effort to develop irrigation infrastructures in Nepal. Chandra Nahar, Juddha Nahar, Jagadisapur (Banganga), Phewa Dam are few examples of the projects developed during that period.
- The minor irrigation program was introduced in the second three- years development plan to provide low-cost irrigation facilities to farmers within short period of time such as small wells, irrigation tanks, reservoir, pumps etc.
- Until mid-१९८०s, irrigation development by the government focused largely on the construction of physical infrastructures of canals and structures, and little attention was given to the effective management of the completed systems.
- The implementation of number of management-oriented projects were carried out such as USAI-funded Irrigation Management Transfer Project (IMTP) in १९८५, Irrigation line of Credit (ILC) in १९८८ financed by the World Bank, Irrigation Sector Project (ISP) IN १९८८ financed by World Bank and Asian Development Bank (ADB) and Irrigation Sector Support Project (ISSP) in १९८९ under ci-financing of UNDP.
- Promulgation of Irrigation Regulations (IR) in १९८९ provided emphasis with water users in all phases of irrigation projects- planning, construction, operation and maintenance.
- Promulgation of Water Resources Act and irrigation Policy in १९९२ with the clear vision of irrigation development, which was later amended in १९९७.
- Now, Irrigation Policy २००४ is in practice.
- Similarly, Irrigation Master Plan १९९०, Agriculture Perspective plan १९९५, Water Resources Strategy २००२ and National Water Plan २००५ guides irrigation development in Nepal.
- Sikta Irrigation Project in Banke, Babai Irrigation Project in Bardiya, Sunkoshi Marine Diversion Project in Kailali are some of the few under construction irrigation project in Nepal.
- Irrigation Projects like Rani Jamara Kulariya Irrigation Project, Bheri-Babai Diversion Multipurpose project are ongoing irrigation projects in Nepal.
- Sunsari-Morang Irrigation system, Kankai Irrigation project, Bagmati irrigation project etc. are the major completed irrigation schemes of Nepal.

b. Crop water requirements calculation by penman methods:

- Crop Water Requirement is the total quantity of water required by crop from the time it is sown to the time it is harvested. It depends upon the type of soil, climate, method of cultivation and rainfall.
- CWR calculation relates with calculation of evapotranspiration by Penman methods and others.

- Evapotranspiration is the combination of evaporation and transpiration.
- Evaporation is the process of liquid converting into vapor, through wind action and solar radiation and returning to the atmosphere.
- Transpiration is the process by which water molecules leaves the body of a living plant and escapes the atmosphere.
- Determination of evapotranspiration by Penman method:
For the area where measured data on the temperature, humidity and sunshine duration or radiation area available, an adaptation of penman method is suggested.

The Modified Penman equation recommended by FAO is:

$$ET_0 = W R_n + (1 - W) f(u) (e_a - e_d)$$

Where :

$$ET_0 = \text{Reference evapotranspiration (mm/day)}$$

$$W = \text{Temperature-related weighting factor}$$

$$R_n = \text{Net radiation expressed as equivalent evaporation (mm/day)}$$

$$f(u) = \text{Windfunction}$$

$$e_a = \text{Saturation vapour pressure (kPa)}$$

$$e_d = \text{Actual vapour pressure (kPa)}$$

$$(e_a - e_d) = \text{Vapour pressure deficit}$$

Wind formula:

For the Modified Penman method:

$$f(u) = 0.27 \left(1 + \frac{u_2}{100} \right)$$

Where :

$$u_2 = \text{Wind speed at 2m height (km/day)}$$

c. Types of irrigation:

Irrigation is an **artificial application of water to agricultural land** in the required quantity and at the required time to supplement natural rainfall for optimum crop growth and yield.

There are several methods of irrigation adopted based on the suitability of local condition such as soil characteristics, topographic conditions, crop rotation, available water flow and others.

Few methods of irrigation are briefly described along with its advantages, limitations and suitability.

i. Surface irrigation:

- In this method, stream of water is diverted from the head of a field into the furrows or borders and allowing it to flow downward.
- Surface irrigation is further classified into:
 - a. Flooding method: entire land surface is covered with water and allowed to stand on the field.
 - b. Contour farming: adapted to hilly areas with steep slopes and quickly falling contours.

- c. Furrow irrigation: common method of irrigation where water is applied in small streams between the rows of the crops, grown on the ridges or furrows.



ii. Sprinkler irrigation:

- Sprinkler irrigation is an improvement over conventional surface irrigation method.
- This method simulates natural rainfall to spread water in the form of rain uniformly over the land surface as needed at uniform pattern and at less rate than the infiltration rate of the soil to avoid the surface runoff.
- Suitability:
 - Sprinkler irrigation is highly suitable for sandy soils, rolling and uneven terrain.
 - It is suitable for closely spaced crops like cereals and fodders.



iii. Drip irrigation:

- Drip irrigation is the method of applying water directly to the plants through a number of low flow rate outlet called emitter or drippers, placed at short interval along with small tubing.
- The network of drippers supply water and fertilizer to the plant roots at the rate of 1 to 2 liter per hour.
- Suitability:
 - This method of irrigation is highly suitable for arid regions and sloppy terrain where the water is scarce.
 - The method is ideal for row crops, orchard and vineyards.



2. Describe the factors to be taken into account while fixing the alignment of irrigation canals.

Answer:

Canal alignment is the route selected for an irrigation canal from the source to the command area. A good alignment ensures economical construction, efficient water delivery, and easy operation and maintenance. Selection of canal alignment is governed mainly by topographic and geological condition along the different routes of cultivable lands.

The factors to be taken into account while fixing the alignment of irrigation canals are listed below:

- i. The irrigation alignment should not pass through the valuable lands, religious places, villages etc. to avoid unnecessary compensation and unwanted conflict.
- ii. The alignment should be short but should not be taken through the area where irrigation is not possible.
- iii. The alignment should be as straight as possible.
- iv. If the curve is unavoidable in the alignment, then it should be provided according to IS 5968-1970.
- v. The alignment should cross the natural stream, drainage etc approximately at right angles.
- vi. The alignment should not involve heavy cutting or banking. It is preferable if balancing depth of cutting and filling is achieved.
- vii. The alignment along the ridge line and watershed line is very good as the watershed canal can irrigate the areas on the both sides.
- viii. The alignment should be such that the maximum area may be irrigated with minimum length of canal.
- ix. The canal alignment should not pass-through marshy lands or water-logged areas.
- x. The alignment should not pass-through sandy soil as percolation loss will be more and the duty of canal will be less.

3. What are the basic differences between Kennedy and Darcy theory? Which theory is used in the design of canal systems in Nepal and why? For a channel to be regime briefly mention the condition that is to be adhered.

Answer:

The basic differences between Kennedy's theory and Darcy's theory:

Kennedy's Theory	Darcy's Theory
1. Kennedy's theory is based on the critical velocity required to prevent silting and scouring in alluvial canals.	1. Darcy's theory is based on the resistance to flow (friction losses) in pipes or channels.
2. The main purpose of Kennedy's theory is to design stable alluvial irrigation canals.	2. The main purpose of Darcy's theory is to calculate head loss due to friction in pipes and channels.
3. Stability depends on maintaining a critical flow velocity.	3. Energy loss depends on friction between flowing water and channel/pipe surface.
4. $(V = m V_0)$, where $(V_0 = 0.55 y^{0.64})$	4. $\left(h_f = f \frac{L}{D} \frac{V^2}{2g}\right)$
5. Applicable mainly to alluvial canals carrying	5. Applicable to pipes and lined or unlined

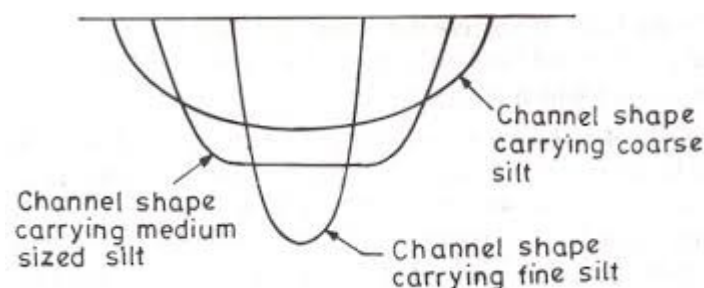
silt-laden water.	channels for friction-loss calculations.
6. Bed slope is determined after satisfying critical velocity.	6. Slope or head loss is obtained directly from friction losses.
7. Does not explicitly account for sediment size or channel roughness.	7. Not suitable alone for designing stable alluvial canals because it ignores sediment behavior.

There are several limitations of Kennedy's theory. They are:

- i. Limitations of Kutter's equation is also incorporated in Kennedy's method.
- ii. No separate equation for slope has been suggested. Kennedy's theory does not indicate as to which channel is best for given discharge.
- iii. He considered only average regime of flow.
- iv. No consideration was made for the silt grade, silt charge, silt concentration and bed load.

Hence, Kennedy's theory is used only in some preliminary designs or checking. For the design of canal systems in Nepal, Lacey's regime theory is used.

- Lacey's regime theory is better and modified method of designing alluvial channel, which considers that the dimensions of bed width, depth and shape attains the state of equilibrium called regime state.
- Lacey also differentiated between initial and final regime condition of channel.
- The initial condition is attained shortly after it is put into operation after construction and the canal begins to adjust its bed slope either by silting or scouring through the bed widths.
- The canal then appears to have attained the stability but is not at the final state of stability and hence known as initial regime condition.
- The continuous action of flowing water overcomes the resistance of the banks and setup a condition such that the canal adjust its complete section and the final regime condition is attained.



Lacey's theory is used for the design of canal systems in Nepal because of the following points:

- i. The theory has been widely adopted in Department of Water Resources and Irrigation (DWRI) practices and irrigation design manuals.
- ii. It provides stable and complete canal design than that of Kennedy's theory.
- iii. Lacey's theory considers the silt size, silt characteristics, wetted perimeter and velocity unlike that of Kennedy's theory.
- iv. The theory is suitable for the alluvial rivers and sediment-laden flows common in Terai region's irrigation systems.

A regime channel is a stable channel whose width, depth and bed slope have undergone modification by silting and scouring and are so adjusted that they have attained equilibrium. For a channel to be regime, the conditions that needs to be adhered:

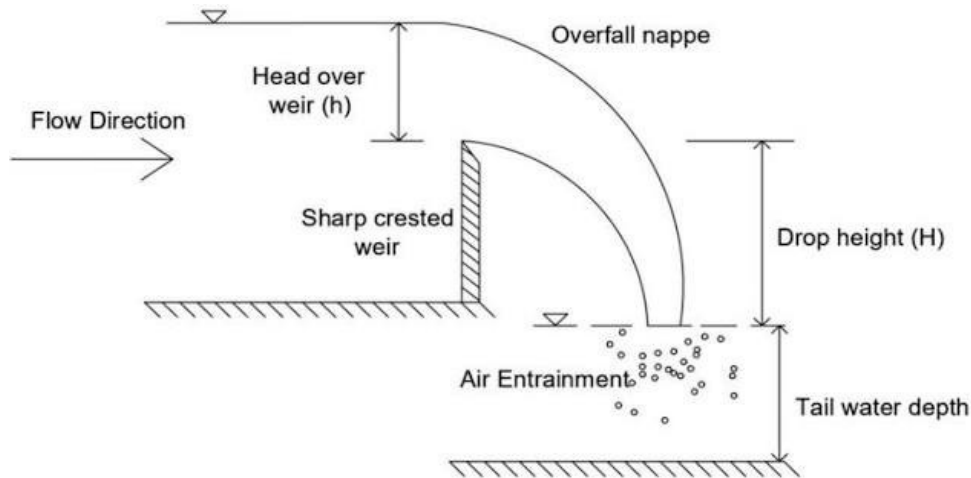
- The channel should carry a constant discharge over a long period.
- The bed and banks should neither **silting (deposition)** nor **scouring (erosion)**.
- The channel bed and banks should consist of **erodible alluvial soil**, allowing adjustment to stable dimensions.
- The channel should have flowed under the same hydraulic conditions for sufficient time to attain a stable section.

- The flow should be steady and approximately uniform throughout the channel.

4. Describe the effect of construction of a weir in the river regime. What are the main causes of weir on permeable foundation and their remedies?

Answer:

Weir is an impervious barrier which is constructed across a river to raise the water level upstream side.



Effects of construction of a weir in the river regime are listed below:

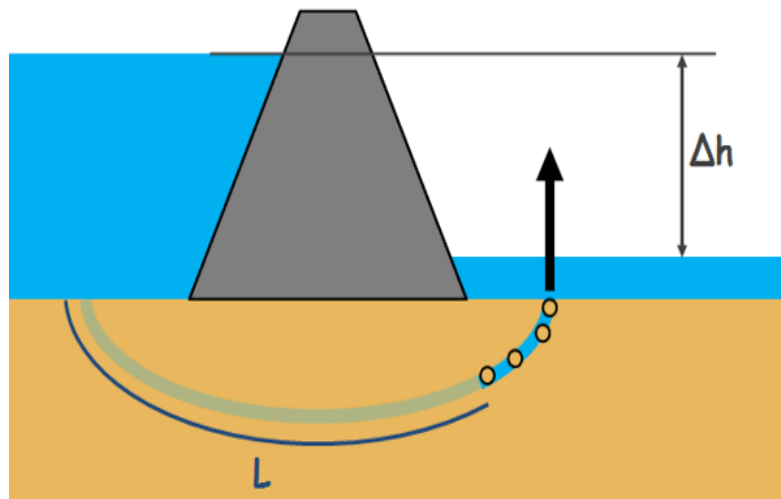
- Weir raises the water level at the upstream sides and facilitates the diversion of the required quantity of water into the canals.
- The raised water level at the upstream reduces the velocity of flow which increases the sedimentation.
- Gradual deposition raises the river bed and reduces the storage capacity of ponded area.
- Weir also interrupts the natural movement of sediment.
- Higher upstream water levels may increase groundwater recharge and raise water table in adjacent areas increasing the risk of waterlogging.
- Weir acts as a barrier to fish migration and movement of other aquatic organisms.
- The raised upstream water level due to construction of weir may also increase flood levels and inundation during high flows.
- Weir also helps to improve irrigation and water supply reliability.

Weirs constructed on **permeable foundations** such as sand, gravel, or alluvium are susceptible to failure due to seepage and erosion. Proper design is essential to ensure stability and safety.

The **main causes of weir on permeable foundation** and their remedies are stated below:

i. Piping (Undermining):

Seepage water carries fine soil particles from beneath the foundation resulting in the formation of underground channels/ tubules leading to collapse of the weir structure.



Remedies:

To prevent piping, we can increase the creep length or provide upstream and downstream cut-offs such as sheet piles. Impervious floor, filters, and drainage arrangements can also be designed and incorporated in the structure.

ii. Scouring at the downstream:

The water with high velocity scours the river bed at downstream exposing the foundation and weakening the structure.

Remedies:

To prevent scouring at downstream, a stilling basin, concrete apron, launching apron, riprap or other energy dissipation works can be designed and constructed.

iii. Scouring at the upstream:

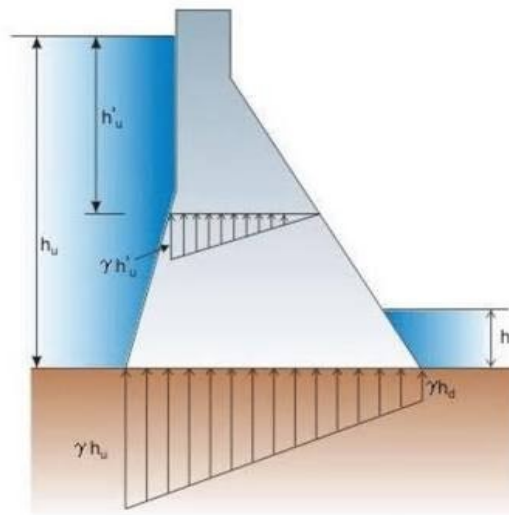
The flow of water at the upstream erodes the river bed and may expose the upstream cutoff.

Remedies:

Upstream launching apron or stone protection can be provided.

iv. Excessive uplift pressure:

The upward pressure created by seepage may lift or crack the floors in the structure.

**Remedies:**

To prevent cracking and lifting of the floor, thicker concrete floor can be constructed by increasing the floor length. Pressure relief drains, cutoff walls can also be provided.

v. Structural failure:

The cracks on the weir, sliding and overturning under the hydraulic loads due to inadequate strength leads to structural failure.

Remedies:

The weir must be designed with adequate factors of safety against sliding, overturning and bearing failure with the use of quality construction materials.

5. What do you mean by hydraulic jump and why is it necessary to be formed in hydraulic structures? For a stable and well-balanced jump which valued of the Froude numbers should to get the anticipation of result. 1) $Fr < 1$ 2) Fr to 1.77 3) Fr

Answer:

- When the depth of flow changes rapidly from a low stage to a high stage, there is abrupt rise of water surface. This local phenomenon is known as hydraulic jump.
- It occurs when the flow changes from supercritical to subcritical.
- Hydraulic jump converts kinetic energy of stream flowing rapidly into potential energy.
- It occurs frequently in a channel below a regulating sluice, at toe of spillway and when the channel bed slope suddenly changes from the steeper to flat slope or where the velocity is high.
- Hydraulic jump forms eddy turbulences in water which causes dissipation of energy.

It is necessary to be formed in hydraulic structures which is explained in the points below:

- i. For the dissipation of excess energy of flowing water over spillway and other hydraulic structure.
- ii. For preventing from possible erosion of downstream floor at structure.
- iii. For increasing the weight of apron floor of hydraulic structure due to increased flow depth.
- iv. For mixing chemicals in water by the help of turbulence created.
- v. For raising the water level in the channel for irrigation.
- vi. For increasing the discharge through a sluice by holding back tail water.

For a **stable and well-balanced hydraulic jump**, the **Froude number (Fr) before the jump should be greater than 1** i.e., the flow must be **supercritical**.

The table below shows the Froude number for different type of hydraulic jump.

Froude Number ((Fr₁))	Type of Hydraulic Jump
(Fr = 1)	No hydraulic jump (critical flow)
(1 < Fr ₁ ≤ 1.7)	Undular jump (weak, unstable)
(1.7 < Fr ₁ ≤ 2.5)	Weak jump
(2.5 < Fr ₁ ≤ 4.5)	Oscillating jump
(4.5 < Fr ₁ ≤ 9)	Stable (steady) hydraulic jump
(Fr ₁ > 9)	Strong jump

A stable and well-balanced hydraulic jump is generally obtained when:

$$4.5 < Fr_1 \leq 9.0$$

6. River flowing through Nepal carries large quantity of sediment load. What measures can be adopted within permissible limits to control the flooding of agricultural land?

Answer:

Nepal is the second richest country in terms of water resources in the world. Water resources such as rivers originates at the Himalayas and carry large amount of sediment due to steep slopes, landslides and erosion. However, due to the fast flow of the rivers in mountain and hilly region, the sediment cannot accumulate at a place. In terai region, the slow flowing rivers give the adequate time for the sediments to settle down. Excess sediment increases river bed level and reduces the capacity of channel which increases the risk of flooding on the agricultural land.

Hence, the measures that can be adopted to control the flooding of agricultural land are as follows:

- i. Embankment/ levees can be constructed along the river bank to prevent floodwater from spreading into agricultural fields.
- ii. River training works such as guide banks, spurs and retaining wall can be constructed to control flooding.
- iii. Sediment control measures such as check dams, silt traps and settling basin can be provided in upstream catchment area.
- iv. Afforestation, terrace farming, slope stabilization can be adopted.
- v. Construction of flood diversion channels, detention basins or reservoirs can also prevent flooding in agricultural land.
- vi. Cultivation and construction of permanent structures in high-risk floodplain areas must be restricted.
- vii. Proper design of hydraulic structures such as bridges, culverts, weirs are done.

7. Do you think ground water irrigation is suitable for the cultivation of crops which requires more water like paddy or sugarcane if not why?

Answer:

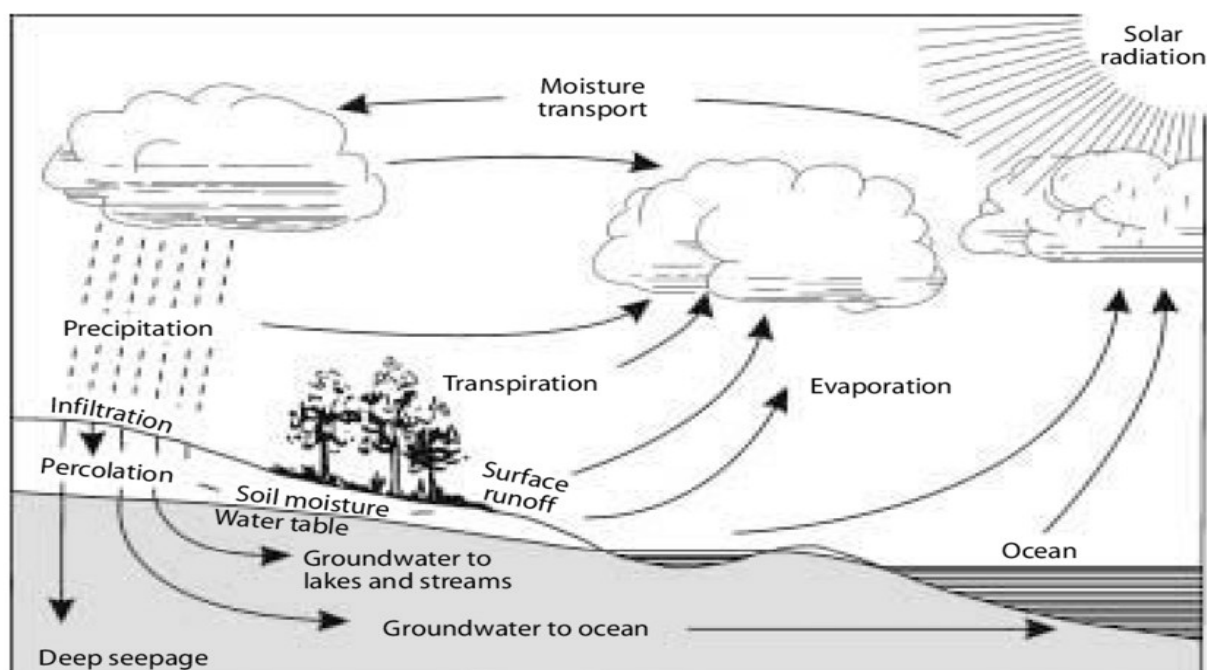
No, groundwater irrigation is generally not suitable as the primary source for cultivating high water-demand crops like paddy and sugarcane, especially on a large scale because of the following points:

- i. Paddy and sugarcane require a large and continuous supply of water throughout their growing season. So, excessive groundwater extraction may not be able to meet this demand sustainably.
- ii. Continuous pumping of groundwater reduces its level which increases the risk of groundwater depletion and drying up of wells.
- iii. Groundwater irrigation requires pumps which increases energy consumption and cost
- iv. If groundwater is withdrawn faster than the rate it naturally charges, the aquifer becomes depleted and threatens future water availability.
- v. Over extraction of groundwater reduces the river baseflow, dry wetlands and cause land subsidence in some areas.
- vi. Surface water sources like rivers generally is more suitable and economical for water intensive crops.

8. Explain with the help of a neat sketch the hydrologic cycle in nature indicating the various phase and describe briefly the basic hydrological data require for planning irrigation projects.

Answer:

- Hydrological cycle is also known as water cycle.
- It is a cycle that involves continuous circulation of water in earth's atmosphere system.
- The cycle has no beginning or end but its process occurs continuously.
- The cycle involves exchange of energy that leads to temperature change.
- The schematic diagram of hydrological cycle is given below:



- The evaporation phase of hydrological cycle purifies the water and replenish the land with fresh water.
- The water from oceans, river, stream, lakes evaporate, move upward and forms clouds by absorbing energy from the sun.
- Water falls back to the sources in the form of rain and also get shifted towards the land due to wind.
- Some portion of rainfall falling on land may get evaporated into the atmosphere while falling.
- Some portion of rainfall may be intercepted by vegetation, leaves, forest floor which later gets evaporated or fall onto the ground.

- Some portion of precipitation falls directly on the ground surface and gets infiltrated into the soil while the other portion runs above and below the ground.
- The infiltrated water may run deep into the crevices and earth reaches to be stored as aquifer by deep percolation.
- Also, the absorbed water by vegetation returns to the atmosphere by transpiration.
- There are two main components of hydrological cycle:
 - i. Transportational components: Precipitation, Evaporation, Transpiration, Infiltration and Runoff
 - ii. Storage components: Soil moisture, ground water storage, storage on land.

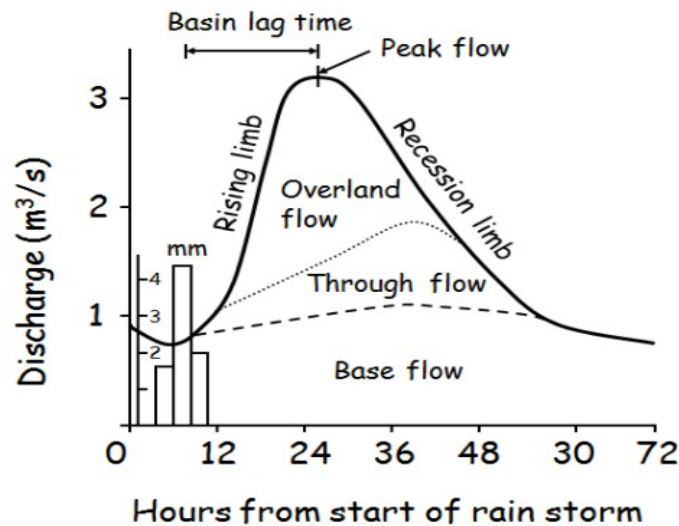
Hydrological data are essential for assessing **water availability**, designing irrigation systems, and ensuring reliable water supply throughout the crop-growing season. The basic hydrological data require for planning irrigation projects are as follow:

- i. Rainfall data:
 - Daily, monthly, and annual rainfall records are collected to determine rainfall intensity, duration, and distribution of rainfall.
 - It is used to estimate **effective rainfall**, crop water requirement, and irrigation demand.
- ii. Runoff data:
 - River or stream discharge records (daily, monthly, annual) are determined to calculate the peak flow, minimum flow, and seasonal variations.
 - It is used to determine the dependable water supply for irrigation.
- iii. Flood data:
 - Flood data is used to determine magnitude, frequency, duration, and return period of floods.
 - Essential for the design of **weirs, barrages, spillways, embankments, and cross-drainage structures**.
- iv. Groundwater data:
 - Groundwater data such as depth of water table, aquifer characteristics and yield are determined.
 - It is used to assess groundwater as a supplementary irrigation source.
- v. Sediment data:
 - Sediment data such as sediment concentration, sediment load, and particle size distribution are to be determined.
 - It is important for the design and maintenance of canals, reservoirs, and diversion structures.
- vi. Characteristics of catchment:
 - Catchment characteristics such as catchment area and shape, topography and drainage pattern, land use and vegetation cover, soil type and infiltration characteristics should be studied.
 - These influence runoff generation and water availability.

9. **a. Define hydrograph. Draw a single peaked hydrograph and indicate its various components.**
b. What are the basic propositions of unit hydrograph theory?

Answer:

a. A hydrograph is the response of a given catchment to a rainfall input. It consists of flow in all the three phases of runoff. It consists of surface runoff, interflow and base flow. Two rainfall in a given catchment produce different hydrographs. Similarly, hydrograph analysis is the study of how stream discharge changes with time after rainfall.



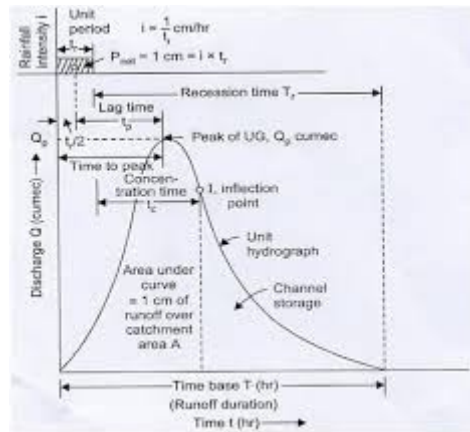
The various components of single peaked hydrograph are listed below:

- i. **Rising limb:**
The steep upward part of the curve after rainfall begins.
The discharge increases as runoff reaches the river.
Indicates the rapid increase in discharge due to rainfall.
- ii. **Peak discharge.**
The highest point on the hydrograph.
this is the maximum river flow during the storm.
- iii. **Lag time.**
The time gap between peak rainfall and peak discharge.
A short lag time usually means a flashier flood response.
- iv. **Recession or falling limb.**
The descending part after the peak, showing discharge gradually returning to normal as runoff decreases.
Represents the gradual decrease in streamflow after rainfall stops.
- v. **Base flow.**
The normal groundwater-fed flow in the channel before and after the storm.
Exists even in the absence of rainfall.
- vi. **Storm flow.**
The extra flow added to the river because of the rainfall event.
- vii. **Direct runoff:**
It is the portion of runoff generated directly by the storm.
It is the area above the base flow line.

b. Unit hydrograph is the direct run of hydrograph resulting from unit depth (1cm) of excess rainfall occurring at the uniform rate for the specific duration of D hours.

Unit hydrograph is the **unit pulse response function** of linear hydrologic system.

The basic propositions of unit hydrograph theory are given below:



The basic assumptions of the unit hydrograph theory in flood forecasting in rivers of Nepal are listed below:

- i. The excess rainfall has constant intensity one by $\frac{1}{D}$ cm/hr for effective storm duration of D hours.
- ii. The excess rainfall is uniformly distributed over the entire catchment area.
- iii. The base time of direct runoff hydrograph (DRH) resulting from excess rainfall of given duration is constant.

The limitations of unit hydrograph theory are stated in the points below:

i. Rainfall Assumptions:

Unit hydrograph assumes the constant intensity of excess rainfall and uniform distribution of excess rainfall over entire catchment which is not possible in practicality as Nepal consists of varying landscape and rugged topography. The assumptions of uniform distribution of rainfall and constant intensity for application of UH fails to resemble the true characteristics of rainfall.

ii. Catchment Area Constraints:

Unit hydrograph is applicable for the catchment having area less than 5000km² square but should not be less than 2km². However, the theory doesn't talk about the large catchments having area greater than 5000 km².

iii. Other forms of precipitation:

It cannot be used if a major portion of the precipitation falls as snow or if a significant part of the basin is snow-covered. It is assumed that precipitation in the basin occurs only in the form of rainfall. However, in mountain regions of Nepal, where snow, glaze, hail are the major forms of precipitation is not considered by the UH theory.

iv. Large Natural Storage:

Catchments with large natural storages such as lakes, swamps, tanks or reservoirs alter the linear relationship between rainfall and runoff which makes the unit hydrograph inaccurate. Hence, this theory limits the presence of large natural storage in the catchment basin.

10. How many types of aquifers exist below the ground surface and what is the main difference between them? Briefly describe what types of well, shallow or deep is feasible in confined and unconfined aquifer?

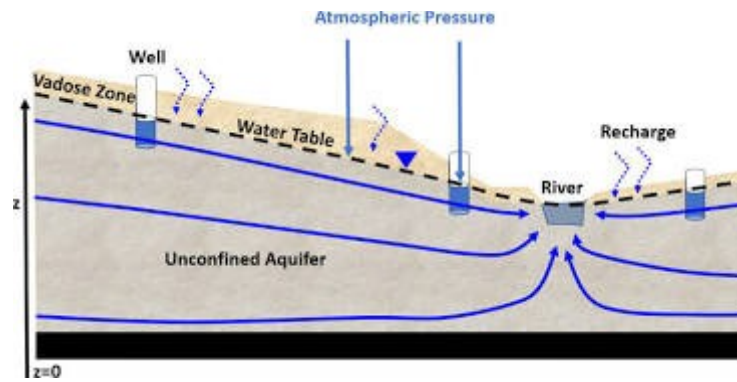
Answer:

There are mainly two types of aquifers that exist below the ground surface. They are:

i. Unconfined aquifers:

The aquifer in which water table forms the upper surface of the zone of saturation is known as unconfined aquifer.

It is also known as water table aquifer or phreatic aquifer or non-artesian aquifer.

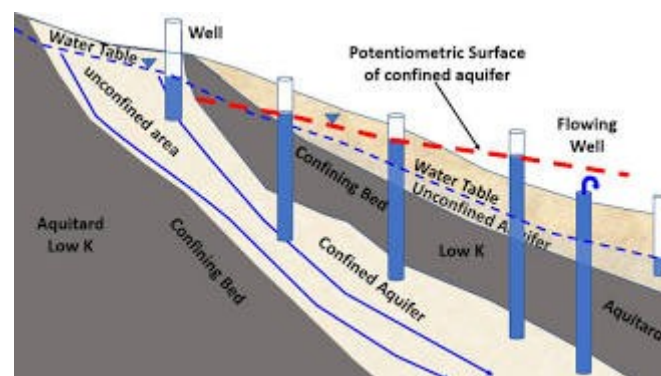


ii. Confined aquifers:

The aquifer in which ground water is confined under pressure greater than atmospheric pressure by overlying relatively impermeable strata is known as confined aquifer.

It is also known as artesian aquifer or pressure aquifer.

Confined aquifer generally occurs at the significant depth below the ground surface.



The differences between confined and unconfined aquifer are as follows:

Unconfined aquifer	Confined aquifer
i. There is no confining upper layer in unconfined aquifer	i. There is an impermeable upper layer in confined aquifer.
ii. Water in unconfined aquifer remains at atmospheric pressure.	ii. Water in confined aquifer is under pressure.
iii. Unconfined aquifer is recharged by direct rainfall and infiltration.	iii. Confined aquifer is recharged from the distant exposed areas.
iv. Storage depends on seasonal recharge and may fluctuate significantly.	iv. Storage in confined aquifer is larger and is more reliable water storage.
v. Examples: shallow sand or gravel aquifer near the ground surface.	v. Examples: deep sandstone aquifer beneath clay layers.

- Unconfined aquifers are best developed using shallow wells because groundwater is close to the surface.
- In unconfined aquifer water table forms the upper surface of the zone of saturation.
- Confined aquifers require deep wells (tube wells) to penetrate the confining layers and access the pressurized groundwater.
- In the confined aquifer, water is confined under pressure greater than atmospheric pressure by overlying relatively impermeable strata.